



p3d: Interactive 3D Visualization of p-Value Surfaces

p3d is an R package for computing and visualizing the **three-dimensional p-value surface** of a single-sample t-test. It expresses the p-value as a function of both effect size (Cohen's d) and sample size (n).

Why p3d?

Traditional p-value visualizations are two-dimensional and show the effect of only one parameter at a time. **p3d** visualizes the p-value simultaneously across effect size and sample size. This answers questions such as:

- How does the same effect size ($d = 0.5$) produce different significance outcomes at $n = 10$ versus $n = 100$?
- Why is the p-value a **continuous measure of evidence**, not a fixed threshold?
- How does sample size moderate the p-value?

Installation

```
# From CRAN (coming soon)
install.packages("p3d")

# Development version from GitHub
```

Links

[Browse source code](#)

[Report a bug](#)

License

[Full license](#)

GPL (≥ 3)

Citation

[Citing p3d](#)

Developers

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Author, maintainer

Dev status

R-CMD-check.yaml | passing



Basic Usage

```
library(p3d)

# Compute a 3D p-value surface
surface <- p_surface(d = seq(-2, 2, 0.1), n = seq(5, 200, 5))

# Interactive 3D visualization (plotly engine)
plot_p_surface(surface, engine = "plotly")

# Static 3D perspective (base R)
plot_p_static(surface)

# 2D p-value curve (for comparison)
curve_30 <- p_curve_2d(d = seq(-2, 2, 0.1), n = 30)
head(curve_30)
```

Main Functions

Function	Purpose
<code>p_surface(d, n, alpha)</code>	Compute the 3D p-value surface
<code>p_curve_2d(d, n, alpha)</code>	Compute a 2D p-value curve (fixed n)
<code>plot_p_surface(surface, engine)</code>	Interactive 3D visualization (rgl / plotly)
<code>plot_p_static(surface)</code>	Static 3D perspective (base R)



$$p(d, n) = 2 \left[1 - F_{t(n-1)}(d\sqrt{n}) \right]$$

where $F_{t(n-1)}$ is the cumulative distribution function of the t-distribution with $n - 1$ degrees of freedom.

The critical effect size at level α is:

$$d_{\text{crit}}(n) = \frac{F_{t(n-1)}^{-1}(1 - \alpha/2)}{\sqrt{n}}$$

Motivation

The p-value is a continuous measure of evidence, yet in practice it is often reduced to a binary decision: “ $p < 0.05$ is significant, $p \geq 0.05$ is not.” This dichotomous thinking is a major contributor to the reproducibility crisis in science.

p3d aims to weaken the visual basis of this dichotomous thinking by visualizing the p-value over a two-dimensional parameter space. The significance boundary is a **curve**, not a **point**.

Performance

`p_surface(.)` computes a 60,100-cell grid in **0.02 seconds** (approximately 3 million p-values per second). Memory usage is under 3 MB.

Contributing

Please use [GitHub Issues](#) for bug reports and feature requests.

